Lam Fung Academy · LF Academy

Primary 5 · 9th Lesson · Student Handout

Fraction multiplication (× fraction) + mixed number processing

Unit 3 · fraction operation (multiplication) · 65 minutes · 1-to-3 Online Lesson

Corresponding textbooks:"New Thinking in Primary School Mathematics (Second Edition)" Volume 5A Unit 3 + Modern Education 5A Unit 11

🔴 **High frequency**The score test must be taken every year, accounting for about 12-18% of Paper 1

Prerequisite knowledge: Class 7-8 (Adding common denominators and different denominators), P4 integers Mixed numbers The sub-test is compulsory every year and accounts for about 12-18% of Paper 1.

Prerequisite knowledge: Lesson 7-8 (Addition of common denominators and different denominators), P4 integers × fractions, HCF and reduction

Our goals:❶ Cross reduction method ❷ Mixed number × fraction (conversion to improper fraction method) ❸ Multiplication of three fractions ❹ Fraction multiplication word problems

Student Name:	Class:	Date:	Time Spent:

I.Warm-Up Questions（total 5 question，5 minutes）

#	Question	Difficulty	Working Space（Show full working）
1	Calculate $3 \times \frac{1}{4} = ?$ (Integer × Fraction, P4 Basics)	Basic	
2	Calculate $\frac{1}{3} \times 2 = ?$ (fraction × integer)	Basic	
3	Convert $1\frac{1}{2}$ to an improper fraction. Then find the improper fraction forms of $1\frac{2}{3}$ and 2 |||SEP|||. $\frac{1}{4}$ Improper fraction form.	Basic	
4	Reduce $\frac{6}{15}$ to the simplest fraction. (HCF(6,15) = ?)	Basic	
5	Find HCF(6, 8), HCF(4, 10), HCF(9, 12). Why is HCF needed for cross reduction?	Advanced	

II.Core Knowledge + Worked Examples

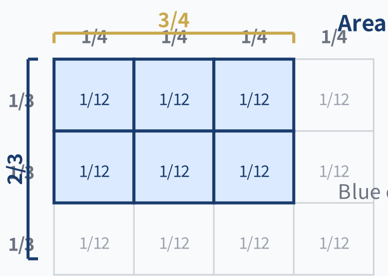
Knowledge Point 1: Fraction × Fraction — Cross Reduction Method 🔴 SSPA

Fraction multiplication rules:

① Numerator × numerator, denominator × denominator: $\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$

② **Cross reduction (reduce first and then multiply) |||SEP|||: Before multiplication, first combine any**numerator**with any**denominator**Use HCF to reduce**③ Advantages: reduce first and then multiply → numbers become smaller → fewer errors → no need for final reduction

④ Note: reduction can only be between numerator and denominator |||SEP|||, not numerator and numerator, denominator and denominator!**between numerator and denominator**, cannot numerator with numerator, denominator with denominator!



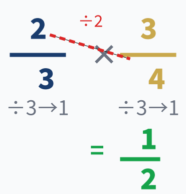
Area model: 2/3 × 3/4 = ?

=

6/12 = 1/2

Blue cells = 6, total cells = 12 → 6/12 = 1/2

Cross reduction indication



□ **Example of trap detonation (the most important demonstration in this class)**

calculate: $\frac{2}{3} \times \frac{3}{4} = ?$

❌ Common mistakes (60% students)	✅ Correct solution (cross reduction)
$\frac{6}{12}$ (not reduced) Directly multiply and then forget to reduce → Not the simplest answer	$\frac{1}{2}$ Cross reduction: numerator 2 and denominator 4 are reduced (HCF=2)→13×32=36=12

🧠 **Tip: "When doing cross-reduction, do it first. The numerator and denominator are about one round. Multiply the rest first and then check. The answer must be the simplest one."**

⚠ **The most frequent error: after multiplying 2/3×3/4=6/12, take it as the answer directly → Forget about reduction! Cross reduction can avoid this error.**

⚠ **The second most frequent error: When approximating a division, the numerator is about the numerator and the denominator is about the denominator → This is wrong! It can only be "crossed" (the numerator is about the denominator).**

Knowledge Point - Worked Examples (mustfirst crosssimplify, againmultiply)

#	Question	Difficulty	Working Space
Example 1	$\frac{1}{2} \times \frac{2}{5}$=? (Hint: Reduction of numerator 2 and denominator 2→$\frac{1}{5}$)	🌱	
Example 2	$\frac{3}{4} \times \frac{2}{9}$=? (Hint: Numerator 3 is approximately the same as the denominator 9, Numerator 2 is approximately the same as the denominator 4)	🌿	

Knowledge Point 2: Mixed Numbers

Standard procedure for multiplication of mixed fractions:

① **Conversion to improper fractions |||SEP|||: Mixed fractions → Improper fractions (integer × denominator + numerator)**Cross reduction |||SEP|||: Reduction between numerator and denominator

② **Numerator × numerator, denominator × denominator**The answer is converted back
③ **: Improper fraction → Mixed fraction (if necessary) + reduction**
④ ⚠ **Absolutely prohibited**: Direct multiplication of mixed fractions! Integer × Integer + Fraction × Fraction ≠ Correct answer!
⚠ **Absolutely prohibited**: Multiply mixed fractions directly! Integer × Integer + Fraction × Fraction ≠ Correct answer!

❌ Wrong way
1 and 1/2 × 1 and 1/3
≠ (1×1) + (1/2×1/3) = 1 and 1/6

→

✅ Correct way
1 and 1/2=3/2 → 3/2 × 1 and 1/3=4/3
= 12/6 = 2

1 and 1/2 = 3/2
1 1 1

1 and 1/3 = 4/3
1 1 1 1

3/2 × 4/3 = 12/6 = 2

①
Convert Improper Fractions
Mixed fraction → Improper fraction

②
cross reduction
The numerator and denominator are reduced using HCF

③
Multiply
Numerator × Numerator Denominator × Denominator

④
Reverse + reduce
False → mixed fraction simplest fraction

Trap comparison question

Calculate: $1\frac{1}{2} \times \frac{1}{3} = ?$

❌ **Error: Multiply mixed fractions directly**

$$1 \times 1 + \frac{1}{2} \times \frac{1}{3} = 1\frac{1}{6}$$

Totally wrong! Mixed numbers must be converted to improper fractions first

✅ **Correct: Convert improper fractions first**

$$\frac{1}{2}$$

$$1\frac{1}{2} = \frac{3}{2} \rightarrow \frac{3}{2} \times \frac{1}{3} = \frac{3}{6} = \frac{1}{2}$$

example

Example 3: $1\frac{1}{2} \times \frac{1}{3} = ?$

example

Example 4: $2\frac{1}{3} \times \frac{3}{4} = ?$

Knowledge point 3: Multiplying three fractions 🔴 SSPA Advanced

Rule of continuous multiplication:

① Multiply all the numerators and all the denominators

② **Comprehensive cross reduction |||SEP|||: Any numerator can be reduced with any denominator (not limited to pairing)**③ Suggestion: Comprehensive reduction first → then multiply → the number becomes super small!

④ Example:

(approximate first, then multiply) $\frac{a}{b} \times \frac{c}{d} \times \frac{e}{f} = \frac{a \times c \times e}{b \times d \times f}$ (Make an appointment before taking a ride)

example

Example 5: $\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4}$ = ? (Hint: Numerator 1, 2, 3 intersects with denominator 2, 3, 4 approximately)

example

Example 6: $\frac{2}{3} \times \frac{3}{5} \times \frac{5}{6}$ = ?

🧠 Formula (continuous multiplication version): "Multiple three brothers in a row, make an appointment together, and then multiply together after the appointment, the answer is so simple."

Knowledge Point 2 + 3 Synchronous practice

#	Question	Difficulty	Working Space
6	$1\frac{1}{4} \times \frac{1}{2} = ?$	🌿	
7	$1\frac{2}{3} \times \frac{1}{5} = ?$	🌿	
8	$\frac{1}{2} \times \frac{1}{4} \times \frac{4}{5} = ?$	🌳	
9	$\frac{1}{3} \times \frac{3}{4} \times \frac{4}{5} = ?$	🌳	

III. Lesson layered synchronization practice (total 15 questions)

Basic layer (total 5 questions, everyone must do)

#	Question	Difficulty	Working Space
10	$\frac{1}{2} \times \frac{1}{3} = ?$	🌱	
11	$\frac{1}{4} \times \frac{1}{2} = ?$	🌱	
12	$\frac{2}{3} \times \frac{1}{2} = ?$	🌱	
13	$\frac{3}{4} \times \frac{1}{3} = ?$	🌱	
14	$\frac{1}{5} \times \frac{5}{6} = ?$	🌱	

Advanced layer (total 5 questions, 🚶🚀 choose do)

#	Question	Difficulty	Working Space
15	$\frac{3}{5} \times \frac{5}{6} = ?$	🌿	
16	$\frac{2}{3} \times \frac{3}{8} = ?$	🌿	
17	$1\frac{1}{4} \times \frac{1}{3} = ?$	🌿	
18	$2\frac{1}{2} \times \frac{1}{5} = ?$	🌿	
19	$\frac{1}{2} \times \frac{1}{3} \times \frac{1}{4} = ?$	🌳	

🌳 challenge layer (total 5 questions, 🚀 choose do, SSPAKiller Questions)

#	Question	Difficulty	Working Space
20	$1\frac{2}{3} \times \frac{3}{5} = ?$	🌳	
21	$2\frac{1}{3} \times 1\frac{1}{2} = ?$	🌳	
22	$\frac{3}{4} \times \frac{8}{9} \times \frac{1}{2} = ?$	🌳	

23	$1\frac{1}{4} \times \frac{2}{3} \times \frac{3}{5} = ?$	🌳	
24	$3\frac{1}{3} \times \frac{3}{5} \times \frac{1}{2} = ?$	⛰	

Knowledge point 4: Fraction multiplication word problems (subject to sub-test word questions) 🔴 SSPA compulsory exam

Four steps to solve the problem:

① **Circle keywords**("... of..." = multiplication! " × divided by

② **Write a complete answer**(Step points will be deducted if you do not write an answer!)

③ **Key words:**"C divided by B" = A ×

④ **Write a complete answer**(Points for steps will be deducted if you don' t write an answer!)

Key words:"A divided by B divided by C" = A × $\frac{C}{B}$

example

Example 7: There are $\frac{2}{3}$ pieces in a cake, and Xiao Ming eats $\frac{1}{2}$ pieces among them. What fraction of the whole cake did Xiao Ming eat? (Hint: "where" $\frac{1}{2}$ = multiplication)

example

Example 8: A rope is $1\frac{1}{2}$ meter long, use |||SEP|||. How many meters were used? (Tip: Convert mixed fractions to improper fractions first and then multiply them) $\frac{3}{5}$. How many meters were used? (Tip: Convert mixed fractions to improper fractions first and then multiply them)

applicationquestionpractice (all must do, column → simplify → calculate → answer sentence)

#	Question	Difficulty	Working Space
25	A box of biscuits weighs $\frac{2}{3}$ kilograms, and Xiaomei ate |||SEP|||. How many kilograms did Xiaomei eat? $\frac{1}{4}$. How many kilograms did Xiaomei eat?	🌿	

26	A bottle of juice contains $\frac{3}{4}$ liters. Xiao Ming poured out |||SEP|||. How many liters were poured? $\frac{2}{3}$ come out. How many liters were poured?	🌿	
27	A rectangle is long $\frac{1}{2}$ meters and its width is long |||SEP|||. How many meters wide is it? $\frac{1}{2}$. How many meters wide is it?	🌿	
28	The elder brother has $1\frac{1}{2}$ kilogram of candy, and he gives $\frac{1}{3}$ to his younger brother. How many kilograms were sent?	🌳	
29	Mom used $\frac{3}{4}$ kilograms of flour to make cakes, of which $\frac{2}{5}$ was used to make cookies. How many kilograms of flour are used to make cookies?	🌳	
30	A ribbon is $2\frac{1}{4}$ meters long, and my sister used |||SEP|||. How much rice did your sister use? (Answer with mixed fractions) $\frac{2}{3}$. How much rice did your sister use? (Answer with mixed fractions)	🌳	

IV. ⛰ Ultimate challenge area (total 3 questions)

#	Question	Difficulty	Working Space
⛰ 1	$\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \frac{4}{5}$ =? (Multiplying four fractions - the ultimate in cross reduction!)	⛰	
⛰ 2	The rectangular area $\frac{2}{3}$ is planted with roses, and the rose part with $\frac{1}{4}$ is a red rose. What fraction of the total area do red roses occupy?	⛰	
⛰ 3	The granary contains $5\frac{1}{3}$ tons of rice. $\frac{1}{4}$ was used in the first week, and $\frac{1}{3}$ was used in the second week (note: is it "the rest" or "all"?). The title says "I used $\frac{1}{3}$ in the second week", which refers to the total amount of $\frac{1}{3}$ used. How many tons were used in two weeks? How many tons are left?	⛰	

V. Class afterhomework

Basic must do questions (total 5 questions, must write cross simplify step)

#	Question	Difficulty	Working Space
H1	$\frac{1}{2} \times \frac{1}{2} = ?$	🌱	
H2	$\frac{2}{3} \times \frac{1}{4} = ?$	🌱	
H3	$\frac{1}{3} \times \frac{3}{5} = ?$	🌱	
H4	$\frac{3}{4} \times \frac{2}{3} = ?$	🌿	
H5	$1\frac{1}{2} \times \frac{1}{4} = ?$	🌿	

Advanced choose doquestion (total 3 questions, 🚀 choose do)

#	Question	Difficulty	Working Space
H6	$1\frac{2}{3} \times \frac{3}{5} = ?$	🌳	
H7	$2\frac{1}{2} \times 1\frac{1}{3} = ?$	🌳	
H8	A rectangular garden $\frac{3}{4}$ is planted with flowers, and the flowers $\frac{1}{3}$ are roses. What percentage of the garden are roses?	🌳	

VI. The Lessoncorecommon errorsummary

✅ **Self-examination in this hall (tick after completion)**

☐ I know the pitfalls of solving each knowledge point

☐ I can complete 🌱basic questions independently

☐ I can challenge 🌿advanced questions

☐ I remember the formula

🎯 **Review of Learning Objectives - After completing this lesson you should be able to:**

☐ Identify all trap types in our hall

☐ Solve 🌱basic questions independently (100% correct)

☐ Challenge 🌿Advanced questions (80%+ correct)

☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	**Cross reduction is done**: $\frac{2}{3} \times \frac{3}{4} = \frac{6}{12}$ when the answer	First make a cross reduction (the numerator 2 and the denominator

		4 are approximately → $\frac{1}{3} \times \frac{3}{2} = \frac{3}{6} = \frac{1}{2}$)
2	**Multiply mixed numbers directly without converting them into improper fractions**	The first step in multiplication with mixed numbers = Convert Improper Fractions! No exceptions!
3	**The numerator is about the numerator, the denominator is about the denominator**	The reduction can only be between the numerator and the denominator (crossover), the numerator cannot be reduced!
4	**The result is unreduced**： $\frac{6}{15}$ When the final answer	Finally the HCF reduction must be checked. Cross-reduction first can avoid this problem.
5	**Improper fractions are not converted to mixed fractions**	When the answer is > 1, the test is required to be converted into a mixed number
6	**When multiplying continuously, only one pair of numerator and denominator will be arranged and then stop |||SEP|||: Only one pair of numerator and denominator will be arranged, and no other appointment will be made**: Only a pair of numerator and denominator are agreed, and no other appointments are made.	When multiplying, all numerators and all denominators can be cross-contracted! Total simplification.
7	**Missing answers to application questions/missing units**	Must write "Answer:..." + unit, otherwise step points will be deducted